

Import libraries for visualization

```
In [ ]: import pandas as pd
import folium
import matplotlib.pyplot as plt
```

```
In [ ]: # Import folium MarkerCluster plugin
from folium.plugins import MarkerCluster
# Import folium MousePosition plugin
from folium.plugins import MousePosition
# Import folium DivIcon plugin
from folium.features import DivIcon
```

Load the dataset collected from earlier

```
In [ ]: filepath = '/Users/frederickhuang/Documents/VSCode/PY/SpaceX Rocket Launches/api_data_with_outcome_labels.csv'
df = pd.read_csv(filepath)
df.head()
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6123.547647	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCSFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

Create map object and dataframe that we'll need to plot markers and locations with.

```
In [ ]: # Start location is NASA Johnson Space Center
nasa_coordinate = [29.55968488583615, -95.0830971930759]
# Create map object starting at NASA JSC
m = folium.Map(location = nasa_coordinate, zoom_start= 5)
```

```
spacex_df = df[['LaunchSite', 'Latitude', 'Longitude', 'Class']]
launch_sites_df = spacex_df.groupby(['LaunchSite'], as_index=False).first()
launch_sites_df = launch_sites_df[['LaunchSite', 'Latitude', 'Longitude']]
launch_sites_df
```

	LaunchSite	Latitude	Longitude
0	CCSFS SLC 40	28.561857	-80.577366
1	KSC LC 39A	28.608058	-80.603956
2	VAFB SLC 4E	34.632093	-120.610829

Create map labels

```
In [ ]: marker_cluster = MarkerCluster().add_to(m)
for lat, long, name in zip(launch_sites_df['Latitude'], launch_sites_df['Longitude'], launch_sites_df['LaunchSite']):
    circle = folium.Circle(location = [lat, long], radius=1000, color='#d35400', fill=True).add_child(folium.Popup(name))

    marker = folium.map.Marker(
        [lat, long],
        # Create an icon as a text label
        icon=DivIcon(
            icon_size=(20,20),
            icon_anchor=(0,0),
            html=<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' % name,
        )
    m.add_child(marker)
    m.add_child(circle)
```



Assign color labels corresponding to successful or failed rocket landing

```
In [ ]: colors = []
for i in range(0, spacex_df.shape[0]):
    if spacex_df['Class'][i] == 1:
        colors.append('green')
    else:
        colors.append('red')
spacex_df['marker_color'] = colors
spacex_df.head()
```

/var/folders/nv/brc1sfvj70n58vqs4_3_5m8h0000gn/T/ipykernel_40087/1377530405.py:7: SettingWithCopyWarning: A value is trying to be set on a copy of a slice from a DataFrame. Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

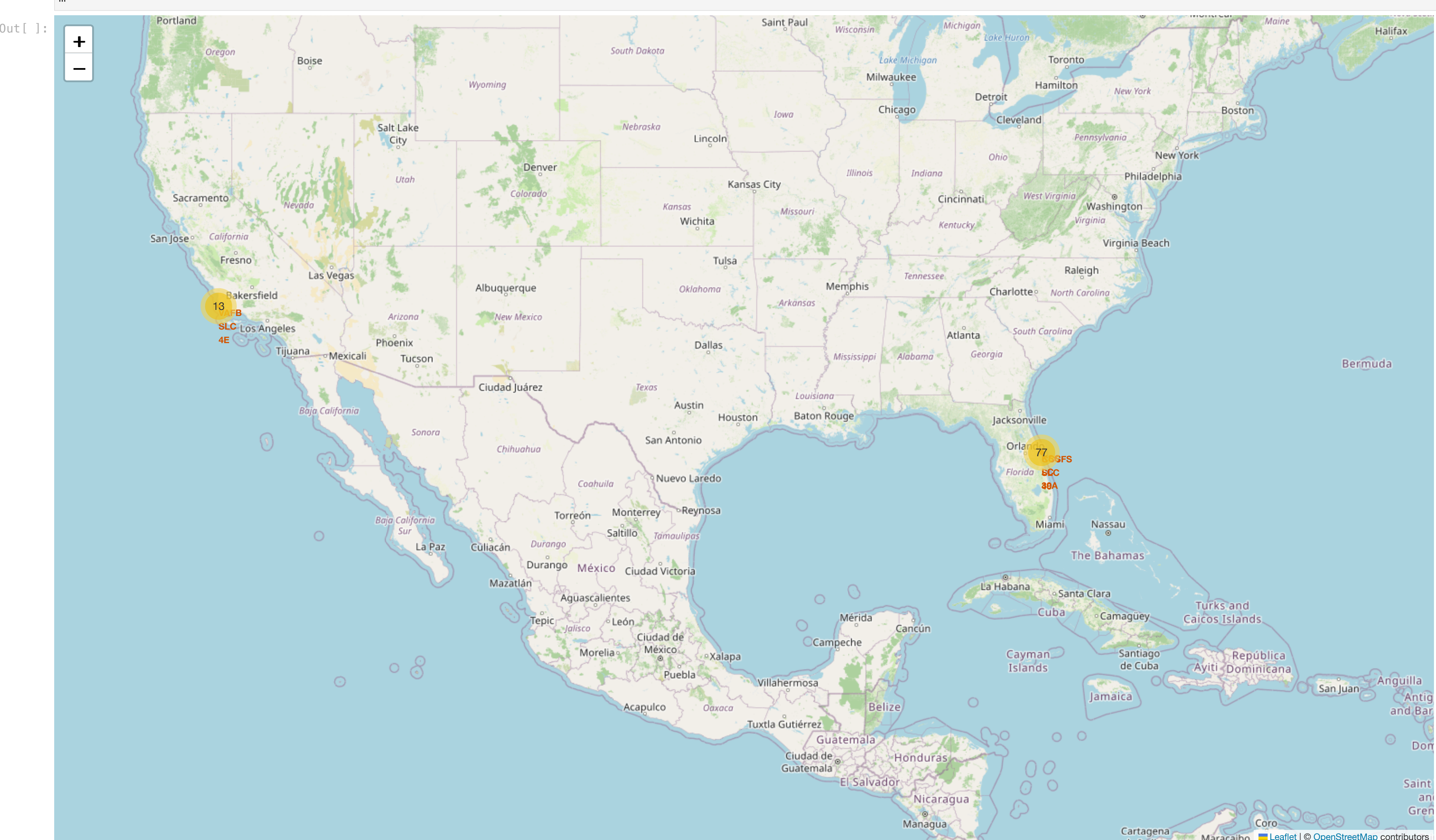
	LaunchSite	Latitude	Longitude	Class	marker_color
0	CCSFS SLC 40	28.561857	-80.577366	0	red
1	CCSFS SLC 40	28.561857	-80.577366	0	red
2	CCSFS SLC 40	28.561857	-80.577366	0	red
3	VAFB SLC 4E	34.632093	-120.610829	0	red
4	CCSFS SLC 40	28.561857	-80.577366	0	red

Plot colored label on map at each of the Falcon 9 launch sites based on launch outcome

```
In [ ]: for lat, long, number, color in zip(spacex_df['Latitude'], spacex_df['Longitude'], spacex_df['Class'], spacex_df['marker_color']):
    marker = folium.Marker(location = [lat, long],

        icon = folium.Icon(color = color)
    )
    marker_cluster.add_child(marker)

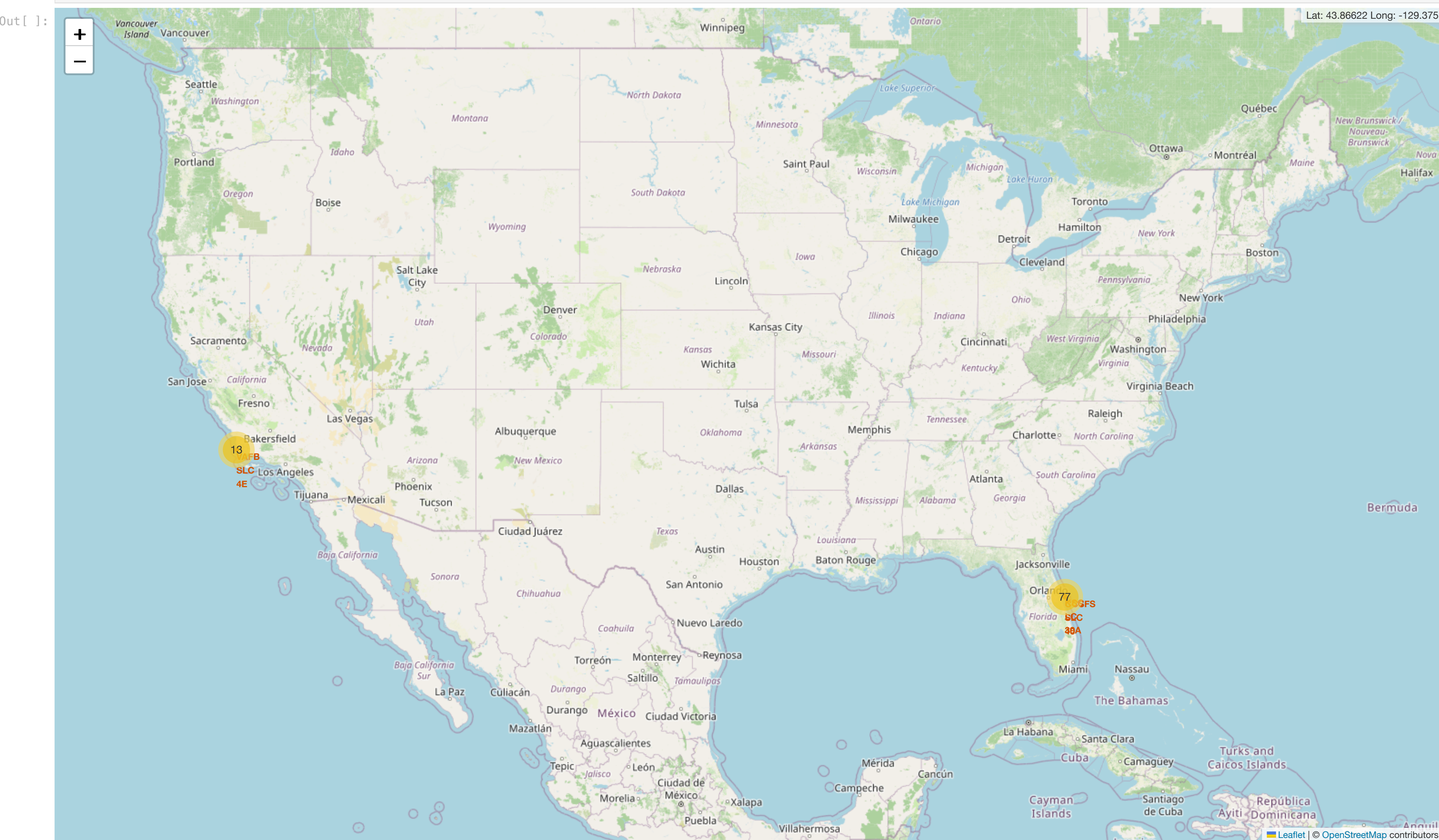
    # print(record)
```



Add mouse position

```
In [ ]: ''' Add Mouse Position to get the coordinate (Lat, Long) for a mouse over on the map
this is mostly for if you want to calculate distances on this map to key geographic features later on
to understand underlying characteristics of our launch sites '''
```

```
formatter = "function(num) {return L.Util.formatNum(num, 5);}"
mouse_position = MousePosition(
    position='topright',
    separator=' Long:',
    empty_string='N/A',
    lng_first=False,
    num_digits=20,
    prefix='Lat:',
    lat_formatter=formatter,
    lng_formatter=formatter,
)
m.add_child(mouse_position)
```



```
In [ ]:
```